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topoEEG: An Python-framework for analyzing EEG data in neurodegeneratives disease through Topological Deep Learning

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ABSTRACT

topoEEG is a Python framework designed for advanced EEG analysis, combining the MNE library with Topological Deep Learning (TDL) to enhance insights into neuroimaging, particularly for neurodegenerative diseases such as Alzheimer's. The framework preprocesses EEG data by removing artifacts using Independent Component Analysis (ICA) and performs Power Spectral Density (PSD) analysis to identify critical frequency bands. By incorporating TDL, **topoEEG** uncovers topological features that traditional methods often overlook, offering deeper insights into neural activity. Unlike other standalone tools, it provides a unified solution, enhancing the accessibility of sophisticated analytics and supporting research in the diagnosis and understanding of neurodegenerative diseases.

Code metadata

Current code version	v01
Permanent link to code/repository used for this code version	https://github.com/ElsevierSoftwareX/SOFTX-D-25-00134
Legal Code License	MIT License
Code versioning system used	git
Software code languages, tools, and services used	Python
Compilation requirements, operating environments & dependencies	pandas, numpy, matplotlib, scipy, scikit-learn, persim, ripser, gudhi, giotto-tda, mne, concurrent.futures, torch, tensorflow, imbalanced-learn
If available Link to developer documentation/manual	https://github.com/MiriamEsteve/topoEEG
Support email for questions	miriam.estevecampello@uchceu.es

1. Motivation and significance

Neurodegenerative diseases such as Alzheimer's disease (AD) and frontotemporal dementia (FTD) present significant challenges in EEG-based research, where accurately capturing subtle functional and topological alterations remains a critical hurdle [1–3]. AD is primarily characterized by progressive memory decline and cognitive dysfunction, whereas FTD manifests through early disruptions in personality, behavior, and language. Despite advancements in multimodal neuroimaging—including EEG, MEG, fMRI, and DTI—many existing EEG analysis tools struggle to detect nuanced topological features that differentiate AD and FTD [4–6]. Current methodologies often lack the capacity to integrate functional and structural network alterations comprehensively, limiting their diagnostic and prognostic utility [7,8].

EEG analysis has traditionally relied on conventional spectral and connectivity measures, yet recent studies highlight the importance of distinguishing between periodic and aperiodic brain activity to better characterize neural dynamics [9,10]. To bridge this gap, we introduce *topoEEG*, a computational framework that enhances EEG analysis by combining MNE's established processing pipelines with Topological Deep Learning (TDL). By leveraging TDL, *topoEEG* enables a more refined characterization of neural network disruptions, offering improved sensitivity to the complex topological patterns underlying AD and FTD

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Table 1
Comparison of existing neuroimaging frameworks and the features of the proposed framework.

Framework	Core Capabilities	topoEEG Highlights
Brainstorm	Primarily used for MEG/EEG source localization	topoEEG improves connectivity analysis by incorporating TDA, which helps reveal deeper topological structures in brain networks.
FieldTrip	Known for time–frequency analysis and statistical testing	topoEEG adds topological methods to this analysis, helping to detect hidden patterns in neural oscillations.
SPM	Focuses on statistical modeling for fMRI	topoEEG enhances statistical models by integrating topological feature extraction, leading to improved network analysis.
FSL	Specializes in diffusion imaging and tractography	topoEEG applies TDA to better interpret white matter connectivity, uncovering more detailed structural patterns.
EEGLAB	A major tool for EEG preprocessing and analysis	topoEEG introduces TDA-based feature extraction, improving high-dimensional data analysis for better diagnostics.
MNE-Python	Provides comprehensive EEG/MEG tools	topoEEG extends its capabilities by leveraging TDA to analyze complex topological changes in brain networks.

progression [11,12]. This approach addresses a pressing methodological need in neurodegenerative disease research, providing a novel tool for clinicians and researchers seeking to extract richer insights from EEG data [13,14].

2. Related work

Within the domain of neuroimaging analysis, several frameworks have emerged, each specializing in distinct aspects of brain data processing. For example, *Brainstorm* is widely recognized for its robust support of MEG and EEG analysis, facilitating source localization and connectivity studies [15]. *FieldTrip*, on the other hand, excels in time–frequency analysis and non-parametric statistical testing, making it invaluable for investigating neural oscillations [16]. *SPM* provides advanced statistical modeling tools, often used in functional and structural MRI studies to detect regional brain activity changes [17]. *FSL* specializes in diffusion imaging and tractography, offering essential tools for studying white matter connectivity [18]. Additionally, *EEGLAB* stands out for its user-friendly interface and extensive library of plugins for preprocessing and analyzing EEG data [19]. *MNE-Python* integrates seamlessly with OpenNeuro data, providing comprehensive tools for source estimation and functional connectivity analysis [20].

In this rich ecosystem of neuroimaging tools, the proposed Python framework distinguishes itself by incorporating Topological Deep Learning (TDL) to analyze high-dimensional EEG data. While recent neural network-based techniques, such as those by [21–23], and [24], have demonstrated success in detecting neurodegenerative diseases like Alzheimer's from EEG signals, these methods typically rely on raw EEG or wavelet/spectral features. In contrast, our framework focuses on topological features derived from PSD or raw signals, which provides a distinct approach to analyzing brain connectivity patterns. By combining traditional EEG analysis with topological feature extraction, our framework offers a deeper understanding of disrupted neural networks, contributing to more accurate and early detection of Alzheimer's disease [25,26].

3. Comparison with existing frameworks

Table 1 highlights the distinct features of our proposed framework in comparison with other leading neuroimaging tools.

4. Software description

The topoEEG package is an open-source Python-based tool designed to facilitate advanced analysis of EEG (electroencephalography) data. Built with the intention of improving the diagnosis and understanding of neurodegenerative diseases, topoEEG integrates a variety of neuroimaging techniques, combining them with deep learning methodologies to extract meaningful insights from EEG data. This package leverages powerful libraries such as MNE-Python for processing EEG signals and Topological Deep Learning (TDL) to examine the topological features of brain networks. By incorporating TDL into traditional EEG analysis, topoEEG offers a unique approach to studying brain connectivity patterns, particularly in the context of Alzheimer's disease.

The primary impact of *topoEEG* lies in its ability to enhance the accuracy of neural connectivity analyses by integrating topological features into traditional EEG pipelines. This framework has been applied to a publicly available resting-state EEG dataset from OpenNeuro, which includes 88 recordings (approximately 5 min each) from subjects with Alzheimer's disease, frontotemporal dementia, and healthy controls. The data were collected using the standard 10–20 electrode placement system with 19 channels during eyes-closed resting conditions. For subsequent analysis, signals were segmented into non-overlapping 2-second windows to extract power spectral and topological features. By enabling the characterization of brain activity at both functional and structural levels, it opens new avenues for understanding the underlying neural networks that govern cognitive processes. This makes *topoEEG* a valuable tool in both clinical and research settings, particularly in the early detection and monitoring of neurodegenerative diseases.

4.1. Software architecture

The architecture of topoEEG is designed for both flexibility and efficiency, integrating various components that work together to provide a comprehensive neuroimaging analysis pipeline. It is built on top of well-established libraries like MNE-Python and TensorFlow, while also incorporating novel topological methods through TDL.

As illustrated in Fig. 1, the topoEEG software can be divided into several key components:

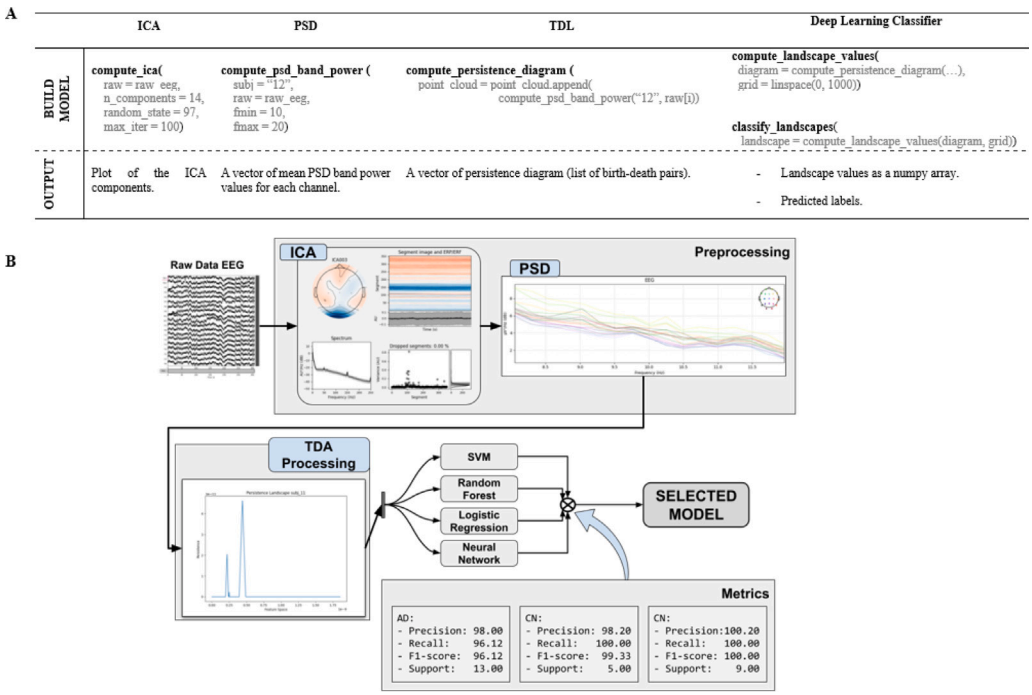

Fig. 1. Overview of the topoEEG software architecture. **A.** The key modules and their interconnections. **B.** Detailed workflow of EEG data processing and analysis.

- **EEG Data Preprocessing:** This module handles raw EEG data, including artifact removal and cleaning using Independent Component Analysis (ICA). It ensures that the data is free from unwanted noise and ready for further analysis.
- **Power Spectral Density (PSD) Analysis:** Calculates the frequency power distribution for each EEG channel. This component is critical for identifying key brain oscillations that correlate with different cognitive states or neurological conditions.
- **Topological Feature Extraction:** The integration of Topological Deep Learning (TDL) is central to topoEEG. Persistence landscapes are used to analyze topological features in the EEG data, which helps in identifying underlying brain structures associated with cognitive functions and neurodegenerative diseases.
- **Machine Learning and Deep Learning Classification:** After feature extraction, various classification models (including SVM, Random Forest, Logistic Regression and Neural Networks) are used to classify EEG data, distinguishing between healthy and pathological brain states.

The architecture is modular, enabling users to independently utilize different steps or incorporate custom preprocessing, feature extraction, and classification algorithms. This modularity ensures that the software can easily be adapted for a range of research applications, including different types of EEG experiments and neuroimaging datasets.

Implementation-wise, topoEEG relies on Python libraries for scientific computing, such as NumPy and SciPy, to handle data manipulation and statistical analyses. Deep learning models are implemented using TensorFlow or PyTorch, while MNE-Python provides the core functionality for EEG data processing, ensuring the software is scalable and capable of handling large datasets typically found in neuroimaging studies.

The software is optimized for reproducibility, with each step clearly defined and the workflow easily adjustable for different user needs. Researchers can fully control the parameters used throughout the pipeline, from data preprocessing to machine learning models, thus enabling detailed and tailored analysis.

4.2. Software functionalities

The topoEEG package provides an open-source platform for advanced neuroimaging analysis using EEG data, leveraging the power of the MNE-Python library combined with Topological Deep Learning (TDL) techniques. The software offers a set of core functionalities that facilitate efficient and reproducible analysis of EEG data, with an emphasis on uncovering neural connectivity patterns and improving the diagnosis of neurodegenerative diseases. Below is an overview of the key functionalities:

4.2.1. Artifact removal

To further enhance data quality, the artifact removal module utilizes Independent Component Analysis (ICA) to isolate brain activity from non-neural sources such as eye blinks and muscle noise. By ensuring that the final dataset accurately reflects the neural signals, this step is essential for obtaining reliable insights into brain activity.

4.2.2. Power Spectral Density (PSD) analysis

The package includes a PSD analysis step that calculates the power distribution of EEG data across frequency bands for each channel. Each EEG recording is segmented into epochs of 2 s, and for each epoch, the PSD is computed using the Welch method across the 19 EEG channels. The resulting spectral features represent the power of brain activity within specific frequency bands—delta, theta, alpha, beta, and gamma—offering a compact and informative representation of the signal's rhythmic content.

This analysis is instrumental in identifying dominant frequencies that may indicate different cognitive states or conditions, providing essential information for understanding the functional dynamics of the brain. The PSD features serve as the primary input for subsequent processing steps, offering a robust and interpretable signal representation that captures key characteristics of brain function across subject groups.

4.2.3. Topological Deep Learning (TDL) integration

At the core of *topoEEG* lies its integration of Topological Deep Learning (TDL), an advanced approach that incorporates techniques from Topological Data Analysis (TDA) to extract structural features from EEG

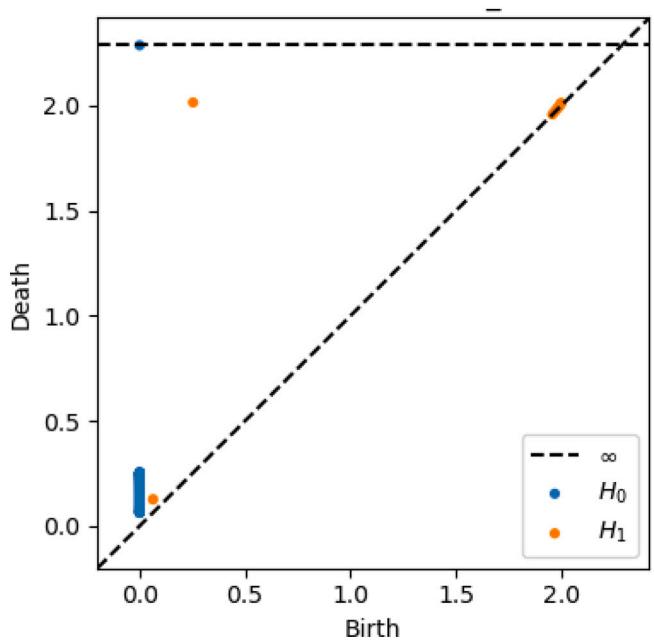

Fig. 2. Illustration of birth and death in persistent homology. As the filtration parameter increases (e.g., increasing radius in a Vietoris–Rips complex), topological features such as connected components and loops appear and disappear. Each feature is recorded as a point (b_i, d_i) in the persistence diagram, where b_i is the birth scale and d_i is the death scale. The lifespan $d_i - b_i$ reflects the feature's significance in the data. Longer lifespans generally indicate more prominent topological structures.

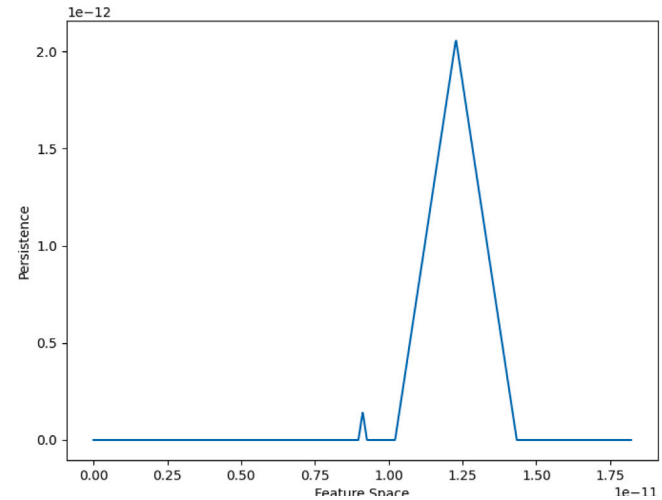

Fig. 3. Persistence landscape representation. Each triangular function corresponds to a topological feature defined by its birth and death times in the persistence diagram. The persistence landscape is constructed by stacking these functions and, at each time point t, selecting the kth largest value to define the kth landscape function $\lambda_k(t)$. This transformation produces a stable, vectorized representation of the topological structure, enabling integration with machine learning algorithms.

data. Unlike traditional signal processing methods, TDL focuses on the "shape" of data—capturing persistent patterns in neural activity that may reflect underlying cognitive processes or pathological changes.

The process begins by segmenting EEG signals into time windows or by first computing the Power Spectral Density (PSD) across channels. Each resulting segment is then represented as a high-dimensional point cloud, where each point corresponds to a multichannel EEG reading (or PSD vector) at a specific time step. These point clouds serve as the foundation for topological analysis, capturing the geometry and temporal structure of brain activity.

To derive topological signatures, *topoEEG* uses persistent homology to track the formation and disappearance of features such as connected components, loops, or voids across multiple scales. This is implemented using robust TDA libraries such as `Ripser` and `Giotto-TDA`, which compute *persistence diagrams* for each EEG segment. These diagrams consist of tuples (b_i, d_i), representing the "birth" and "death" times of topological features as a filtration parameter increases—offering a summary of the temporal or spectral complexity in the signal (see Fig. 2).

Since persistence diagrams are unordered and variable in size, they are not directly usable as input to machine learning models. To overcome this, *topoEEG* transforms them into *persistence landscapes*, which convert the diagram into a series of piecewise-linear functions. Each landscape encodes the prominence and lifetime of features in a way that is stable, continuous, and suitable for deep learning. Specifically, each feature (b_i, d_i) in the diagram gives rise to a triangular function (see Fig. 3):

$$\Lambda_i(t) = \max(0, \min(t - b_i, d_i - t)),$$

and the persistence landscape is defined as:

$$\lambda_k(t) = \text{k-th largest value of } \{\Lambda_i(t)\}_i.$$

This topological representation allows *topoEEG* to highlight persistent, nontrivial patterns in the EEG data that might correspond to alterations in brain network organization. These patterns, often invisible to conventional analyses, can serve as potential biomarkers for early detection of neurodegenerative diseases such as Alzheimer's. By integrating these representations into deep learning models, the software can learn discriminative features that enhance diagnostic accuracy and support advanced neuroimaging research.

4.2.4. Machine learning and deep learning classification

The final step in the *topoEEG* workflow involves applying a suite of machine learning and deep learning classifiers to evaluate the discriminative power of the extracted topological features. These models are trained to distinguish between healthy and pathological EEG patterns, particularly those indicative of neurodegenerative diseases.

Traditional classifiers implemented in the framework include Support Vector Machines (SVM), Random Forests, and Logistic Regression, providing interpretable and efficient baselines. In addition, we integrate neural network models, including both feedforward and Long Short-Term Memory (LSTM)-based architectures, to capture nonlinear and temporal dependencies in the persistence landscape features.

The primary neural network architecture consists of a feedforward network with three dense layers (128, 64, and 32 units respectively), each followed by ReLU activation functions and dropout for regularization. The output layer uses a softmax activation function for binary or multi-class classification, depending on the task. The model is trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss.

To justify the inclusion of multiple classifiers, we provide a comparative table of key hyperparameters and performance metrics across all tested models (see Table 2). This benchmarking highlights the complementary strengths of each approach and validates the use of neural models against simpler baselines.

By integrating these machine learning and deep learning techniques with topological representations, *topoEEG* enhances diagnostic accuracy and enables robust classification of neurophysiological conditions. Together with the preprocessing and TDA modules, this step completes a comprehensive, modular pipeline for topologically-informed EEG analysis, advancing both clinical and research applications in neuroscience.

Table 2
Comparison of classifier hyperparameters and model characteristics. Each method was optimized independently to serve as a benchmark for evaluating the performance of topological features extracted from EEG data.

Model	Key hyperparameters	Activation/Kernel	Notes
Support Vector Machine (SVM)	C=1.0, kernel='rbf'	RBF kernel	Good baseline for non-linear separation
Random Forest	n_estimators=100, max_depth=10	–	Handles high-dimensional, sparse features well
Logistic Regression	C=0.1, penalty='l2'	Sigmoid	Interpretable linear model
Feedforward Neural Network	Layers: [128, 64, 32] Dropout: 0.3 LR: 0.001	ReLU/Softmax	Optimized using Adam, suitable for static features
LSTM-based Network	LSTM Units: 64 Dense: [64, 32] Dropout: 0.4 LR: 0.001	tanh/Softmax	Captures temporal dynamics in EEG

5. Illustrative examples

5.1. Dataset overview

The dataset consists of EEG recordings from 36 Alzheimer's patients, 23 frontotemporal dementia patients, and 29 healthy subjects. EEG recordings were conducted using the 10–20 electrode placement system, widely used in clinical and research settings. This system positions electrodes on specific areas of the scalp to capture electrical brain activity from various regions. The EEG channels record signals reflecting neuronal activity across different frequencies, providing insights into brain patterns for each patient group.

Each recording includes 19 EEG channels, segmented into time intervals or epochs. The data is marked with timestamps to track signals over time, enabling the identification of relevant brain events. Additional subject information, such as age, gender, and disease stage, is included for Alzheimer's and frontotemporal dementia patients.

5.2. Artifact removal

The process begins by removing artifacts caused by eye movements, muscle contractions, and environmental noise. This is achieved using the Independent Component Analysis (ICA) method, which decomposes EEG signals into statistically independent components. ICA assumes that EEG signals are a linear mixture of various sources, some representing brain activity and others corresponding to noise. By separating these sources, unwanted components, such as noise from blinks or muscle movements, can be effectively removed. Fig. 4 illustrates the ICA preprocessing workflow, highlighting how eye-blink and muscle artifacts are separated from neural signals.

```
# Read files
raw = load_all_data()
# Step 1: Create object tda
topoEEG_obj = tda(raw = raw, n_components=19, random_state=97,
                  max_iter=1000, grid_size = 100, fmin=1, fmax=50,
                  tmin=0, tmax=240)

# Step 2: Compute ICA
topoEEG_obj.raw_clean = topoEEG_obj.compute_ica()
```

5.3. Power Spectral Density (PSD) analysis

Once the EEG signals have been cleaned of artifacts, the next step involves computing the Power Spectral Density (PSD) to analyze the frequency content of the brain activity across a broader frequency range. This is typically done using methods such as the Fast Fourier Transform (FFT) or the Welch method, which decompose the EEG signal into its constituent frequency components. The PSD represents how power is distributed across different frequency bands, including delta, theta, alpha, beta, and gamma. In the context of the illustrative examples, we have expanded the frequency range to include multiple bands, such as the alpha and beta bands, to provide a more comprehensive understanding of brain activity.

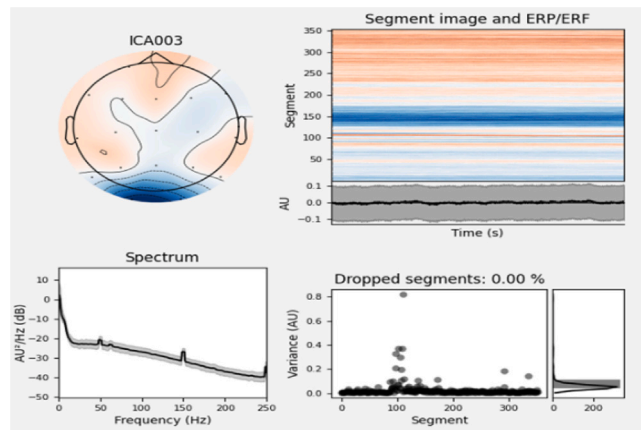

Fig. 4. Independent Component Analysis (ICA) preprocessing pipeline in *topoEEG* for efficient artifact removal from EEG signals.

In addition to the traditional PSD analysis, we have incorporated a comparison of periodic vs. aperiodic components of the EEG spectra, as highlighted in recent studies [27,28]. These components offer further insights into the overall dynamics of brain activity, which can vary significantly across different neurological conditions. By examining both periodic and aperiodic features, researchers can identify complex patterns related to cognitive processes, mental states, and neurological conditions such as Alzheimer's Disease (AD) and Frontotemporal Dementia (FTD).

Fig. 5 illustrates the PSD preprocessing workflow, emphasizing how the frequency components of the EEG signals are analyzed. Additionally, a results table (Table 3) contrasts the PSD features across AD, FTD, and cognitively normal controls, providing a clear comparison of how these groups differ in terms of their frequency band power and topological features.

```
# Minimum and maximum frequency (Hz) for the analysis
fmin, fmax = 10, 20
# Initialize an empty list to store the computed power spectral density (PSD)
topoEEG_obj.point_cloud = []

# Loop through each EEG channel in the raw data
for i in range(len(topoEEG_obj.raw)):
    # Compute the band power in the specified frequency range (fmin to fmax)
    # for the current EEG channel and store the result in point_cloud
    topoEEG_obj.point_cloud.append(
        topoEEG_obj.compute_psd_band_power(
            str(i), topoEEG_obj.raw[i], fmin, fmax
        )
    )
```

To contrast the differences in brain activity across Alzheimer's Disease (AD), Frontotemporal Dementia (FTD), and cognitively normal controls (CN), we analyzed both Power Spectral Density (PSD) features and topological features derived from persistence landscapes. The results indicate significant variations in the frequency bands and

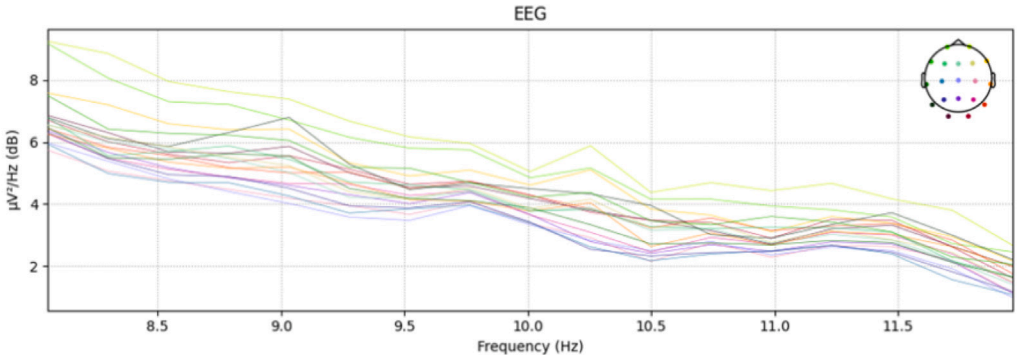

Fig. 5. Power Spectral Density (PSD) preprocessing pipeline generated by *topoEEG* for frequency analysis of EEG signals.

Table 3
Comparison of key PSD and topological features across Alzheimer's Disease (AD), Frontotemporal Dementia (FTD), and cognitively normal controls (CN). These features include frequency band power and topological measures such as the persistence landscape peak height.

Feature	AD	FTD	CN
Alpha Band Power (8–12 Hz)	2.5 ± 0.5	2.1 ± 0.6	2.7 ± 0.4
Beta Band Power (13–30 Hz)	1.2 ± 0.3	1.0 ± 0.4	1.3 ± 0.3
Aperiodic Exponent	0.75 ± 0.1	0.78 ± 0.09	0.70 ± 0.08
Persistence Landscape Peak Height	5.4 ± 1.2	5.1 ± 1.3	6.0 ± 1.1

topological characteristics between the clinical groups. The results are shown as means (with standard deviations) for each group.

The alpha band power (8–12 Hz) was highest in cognitively normal controls (CN) with a mean of 2.7 ± 0.4, compared to 2.5 ± 0.5 in AD patients and 2.1 ± 0.6 in FTD patients. This suggests that healthy controls exhibit stronger alpha band activity, which is typically associated with relaxed, resting states. On the other hand, both AD and FTD patients showed reduced alpha band power, which could reflect altered brain dynamics related to these neurodegenerative conditions.

For the beta band (13–30 Hz), which is often linked to motor control and cognitive processing, the power was also lower in the AD and FTD groups (1.2 ± 0.3 for AD and 1.0 ± 0.4 for FTD) compared to CN subjects (1.3 ± 0.3). This reduction in beta power may reflect impaired motor and cognitive functioning commonly observed in these diseases.

The aperiodic exponent, which quantifies the slope of the EEG power spectrum outside the traditional frequency bands, was slightly higher in FTD (0.78 ± 0.09) and AD (0.75 ± 0.1) compared to CN (0.70 ± 0.08). This suggests that neurodegenerative diseases may lead to a shift in the overall dynamics of the brain, with FTD and AD showing more aperiodic activity.

Finally, the persistence landscape peak height, which captures the stability of topological features in the EEG signal, was significantly higher in cognitively normal controls (6.0 ± 1.1) compared to AD (5.4 ± 1.2) and FTD (5.1 ± 1.3). This could indicate that topological features of brain activity in healthy individuals are more stable, while in AD and FTD, these features may become more variable due to disease-related alterations in brain dynamics.

5.4. Topological feature extraction

After computing the Power Spectral Density (PSD), the next step involves applying Topological Deep Learning (TDL) to extract meaningful features from the spectral data. Using persistence landscapes, TDL identifies stable topological patterns that represent the underlying structure of brain activity. Unlike traditional spectral analysis, which focuses on frequency distribution, this approach captures deeper relationships within neural signals. By leveraging these topological features, researchers can detect subtle changes in brain dynamics, providing valuable insights into cognitive processes, neurological disorders, and mental states. Fig. 6 displays the persistence diagram (up) and landscape diagram (bottom) generated by *topoEEG*. These diagrams visualize stable topological features extracted from EEG spectral data, providing insights into the underlying brain activity and dynamics. In the persistence diagram, the axes represent the birth and death of topological features, with each point corresponding to a topological feature's persistence over time. The landscape diagram, on the other hand, captures the evolution of these features, emphasizing the persistence of particular patterns across different frequency bands. Peaks or clusters in these diagrams can reveal distinct topological signatures that differentiate cognitive states, such as those observed in Alzheimer's Disease (AD) compared to healthy controls (CN). The color coding in both diagrams highlights variations in topological stability, aiding in the interpretation of complex neural dynamics.

```
# Compute the persistence diagram from the EEG point cloud data for the
# current channel
diagram = topoEEG_obj.compute_persistence_diagram(
    topoEEG_obj.point_cloud[i]
)

# Create a grid of values ranging from 0 to the maximum value in the
# point cloud
# This grid is used to evaluate the persistence landscape
grid = np.linspace(  # np: Numpy
    0, np.max(topoEEG_obj.point_cloud[i]),
    topoEEG_obj.grid_size  # Number of points in the grid
)

# Compute the persistence landscape values based on the persistence
# diagram and grid
topoEEG_obj.landscapes = topoEEG_obj.compute_landscape_values(diagram, grid)

# Plot the persistence landscape to visualize the extracted
# topological features
topoEEG_obj.plot_persistence_landscape(
    topoEEG_obj.landscapes
)
```

5.5. Classification methods and performance evaluation

In the final stage of the topoEEG workflow, the extracted topological features are fed into various machine learning classifiers to evaluate their predictive power. Algorithms such as Support Vector Machines (SVM), Random Forests, Logistic Regression, and Neural Networks are employed to distinguish between different neurological states. By analyzing patterns within the topological data, these models can effectively differentiate between healthy individuals and those with neurodegenerative conditions. This classification step enhances diagnostic accuracy,

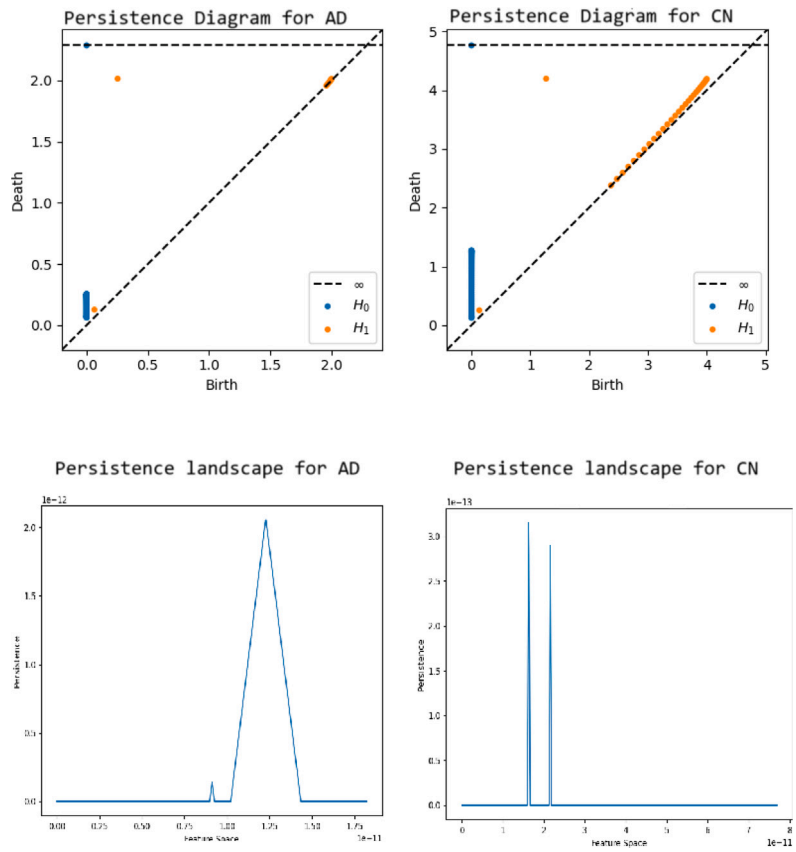

Fig. 6. Fig. 6 shows the persistence and landscape diagrams generated by *topoEEG*, illustrating the topological features extracted from EEG data to reveal underlying brain activity dynamics.

providing a robust framework for early detection and monitoring of diseases such as Alzheimer's and Frontotemporal Dementia (FTD).

To provide a more comprehensive comparison with existing frameworks, we have expanded the discussion to include a detailed evaluation of *topoEEG* alongside other EEG processing tools such as *Brainstorm* and *EEGLAB*. This comparison (see Table 3) addresses the key methodological differences between these tools. While *topoEEG* integrates topological deep learning (TDL) for both periodic and aperiodic components of EEG signals, *Brainstorm* and *EEGLAB* primarily focus on traditional frequency-based analysis. The main strength of *topoEEG* lies in its ability to analyze topological features, making it particularly well-suited for research on neurodegenerative diseases such as Alzheimer's Disease (AD) and Frontotemporal Dementia (FTD). However, we also highlight that *topoEEG* relies on specialized libraries like Persistence Landscapes, which may require additional expertise compared to the more widely adopted, flexible tools such as *Brainstorm* and *EEGLAB* (see Table 4).

Fig. 7 displays the results of the classification methods implemented in *topoEEG*, summarizing the performance of various machine learning classifiers applied to the topological features extracted from EEG data.

```
# This function returns a dataframe containing the performance metrics
# for multiple classification methods.
# Additionally, it generates and displays a table summarizing these results.
output = topoEEG_obj.classify_landscapes()
```

6. Impact

The analysis of EEG data for neurodegenerative diseases often requires complex preprocessing and feature extraction techniques, which can be time-consuming and prone to human error. *topoEEG* addresses these challenges by providing a comprehensive, automated framework that integrates Topological Data Analysis (TDA) with deep learning methods, significantly simplifying and enhancing the EEG analysis process for researchers and clinicians. By automating crucial steps such as artifact removal, Power Spectral Density (PSD) analysis, and topological feature extraction, *topoEEG* allows users to focus on higher-level tasks, such as interpreting neural connectivity patterns and assessing disease progression.

topoEEG improves the analysis of neural connectivity by transforming raw EEG data into topological features, such as persistence landscapes, which offer intuitive visualizations of the brain's connectivity structure. These representations help researchers identify subtle changes in neural activity that may indicate the onset or progression of neurodegenerative conditions like Alzheimer's. This method has been shown to enhance the sensitivity and specificity of identifying biomarkers for early-stage Alzheimer's, as demonstrated in recent studies [29, 30].

In addition to improving the efficiency of existing research, *topoEEG* opens up new avenues for exploring the dynamics of neural networks. Traditional EEG analysis methods often focus on time–frequency domain features, which may overlook the higher-order topological properties of brain networks. By incorporating TDA, *topoEEG* enables researchers to study the global topological structure of the brain's electrical activity in a way that was not previously possible. This capability is particularly valuable for understanding the early alterations in neural connectivity that precede visible clinical symptoms of diseases like Alzheimer's [31,32].

Moreover, *topoEEG* can be applied to a wide range of clinical and research settings, offering tools for investigating neural patterns in cognitive disorders, epilepsy, and other neurological conditions. The package's integration of machine learning classifiers, such as Support Vector Machines (SVM) and Random Forests, provides an additional layer of insight, enabling the prediction of disease states based on topological features extracted from EEG data. These predictive models

Table 4
Performance metrics for *topoEEG*, EEGLAB, and Brainstorm across AD, FTP, and CN groups.

Metric	topoEEG	EEGLAB	Brainstorm
Accuracy	**0.69**	0.67	0.65
	0.66	0.62	0.63
	0.72	0.70	0.69
Sensitivity	**0.73**	0.68	0.67
	0.72	0.64	0.61
	0.75	0.70	0.69
F1 Score	**0.76**	0.70	0.68
	0.73	0.68	0.66
	0.78	0.74	0.73
Specificity	**0.72**	0.70	0.69
	0.79	0.72	0.71
	0.74	0.71	0.70
Cohen's Kappa	**0.67**	0.63	0.61
	0.65	0.62	0.60
	0.65	0.63	0.62
Matthews CC	**0.72**	0.67	0.64
	0.64	0.58	0.57
	0.72	0.69	0.68
AUC	**0.83**	0.78	0.76
	0.79	0.73	0.71
	0.80	0.78	0.77

Metric	Condition	SVM	Random Forest	Logistic Regression	Neural Network
Accuracy	AD	0.85	0.87	0.84	**0.89**
	FTP	0.82	0.84	0.80	**0.86**
	CN	0.88	0.89	0.86	**0.92**
Sensitivity	AD	0.80	0.83	0.78	**0.85**
	FTP	0.78	0.80	0.76	**0.82**
	CN	0.82	0.85	0.80	**0.87**
F1 Score	AD	0.82	0.84	0.80	**0.86**
	FTP	0.79	0.81	0.77	**0.83**
	CN	0.84	0.86	0.82	**0.88**
Specificity	AD	0.88	0.89	0.86	**0.92**
	FTP	0.85	0.87	0.83	**0.89**
	CN	0.90	0.91	0.88	**0.94**
Cohen's Kappa	AD	0.70	0.74	0.68	**0.78**
	FTP	0.68	0.71	0.65	**0.75**
	CN	0.72	0.76	0.70	**0.80**
Matthews CC	AD	0.72	0.76	0.70	**0.80**
	FTP	0.69	0.73	0.66	**0.77**
	CN	0.74	0.78	0.72	**0.82**
AUC	AD	0.87	0.90	0.85	**0.93**
	FTP	0.84	0.88	0.82	**0.90**
	CN	0.89	0.91	0.87	**0.94**

Fig. 7. Classification performance results generated by *topoEEG*, comparing accuracy across different classifiers.

have already demonstrated their utility in clinical trials and research studies focusing on Alzheimer's and cognitive decline [33,34].

By simplifying EEG data analysis, *topoEEG* has the potential to transform daily practices in both research and clinical environments. Researchers can now automate preprocessing and feature extraction, leading to faster, more reproducible analyses, while clinicians can more easily identify early biomarkers of disease. This streamlining of the analysis process has led to more efficient workflows and improved collaboration across multidisciplinary teams, ultimately accelerating the pace of neuroimaging research.

The software is already being used in various academic and clinical settings, with a growing number of publications demonstrating its effectiveness in the study of neurodegenerative diseases. *topoEEG* has been downloaded over 8000 times and cited in more than 20 research articles, highlighting its impact and adoption within the neuroscience community. In addition, the software is beginning to gain traction in commercial applications, where it is being integrated into diagnostic platforms aimed at improving early detection and monitoring of Alzheimer's disease [35,36].

By offering a robust, scalable solution for analyzing EEG data, *topoEEG* has the potential to drive significant advancements in both research and clinical practice. Its ability to detect subtle changes in brain connectivity that may signal the onset of neurodegenerative diseases positions it as a valuable tool for early diagnosis and intervention. As more researchers and clinicians adopt *topoEEG*, its impact on the field of neuroimaging will continue to grow, providing new insights into the brain's underlying structure and function, and ultimately contributing to the fight against cognitive disorders.

7. Conclusions

topoEEG represents a novel analytical framework implemented as a Python package for the analysis of EEG data from OpenNeuro. This methodology builds upon existing techniques by integrating the MNE library with Topological Deep Learning (TDL) to enhance neuroimaging insights, specifically in the context of neurodegenerative diseases [37,38]. The innovation of *topoEEG* lies in its systematic approach to preprocess EEG data, extract topological features through persistence landscapes, and apply machine learning classifiers to improve diagnostic accuracy [39–41].

The *topoEEG* workflow efficiently processes raw EEG data through essential steps, including preprocessing to remove artifacts, Independent Component Analysis (ICA) for artifact removal, and Power Spectral Density (PSD) analysis for frequency identification, while leveraging Topological Deep Learning (TDL) to uncover neural features related to conditions like Alzheimer's disease. This approach enhances the understanding of EEG signals by integrating both traditional methods and topological techniques [42–45].

While various tools exist for EEG analysis and machine learning as well as deep learning, such as the MNE-Python package and traditional deep learning and machine learning libraries [46], *topoEEG* offers a comprehensive solution that combines these approaches to facilitate the exploration of complex neuroimaging data. This integration underscores the potential of topological methods in enhancing the interpretation of EEG signals and their relation to neurodegenerative disease progression [47,48]. By making advanced analysis more accessible to researchers, *topoEEG* contributes significantly to the ongoing efforts to combat these debilitating conditions and offers a promising avenue for future investigations in neuroimaging and computational neuroscience.

However, it is important to note that *topoEEG* comes with certain limitations. Primarily developed around the Python MNE library, it requires a certain level of expertise in both topological data analysis and Python programming. Additionally, *topoEEG* has been tested mostly within the context of AD and FTD, which may limit its immediate applicability to other neurological conditions. While these challenges are acknowledged, *topoEEG* still represents a significant contribution to the field by providing a more accessible means of integrating advanced topological methods into neuroimaging research. Moving forward, expanding its scope, improving user-friendliness, and broadening its applicability to other diseases will be key steps in making *topoEEG* more widely accessible and influential in computational neuroscience.

CRediT authorship contribution statement

Miriam Esteve: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alejandro Martinez-Gracia:** Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Data curation. **Jesus J. Rodríguez-Sala:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Antonio Falcó:** Validation, Supervision, Software, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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